Recent Augmented Reality Applications

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Overview

- Augmented Reality
- Computer Vision

Recent Augmented Reality Applications

Sixth Sense

Sixth Sense

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Sixth Sense in 21st Century

Dr. Pattie Maes,
Massachusetts Institute of Technology (MIT), United States

Broadcast in TED

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TED TALKS

Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- "*The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques*"..from MIT Media Lab

Augmented Reality Guitar

Augmented Reality Guitar

Dr. Chutisant Kerdvibulvech,
Keio University, Japan

Broadcast in Discovery Channel

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MR for Military Application

Mixed Reality for Military Application

Dr. Adrian Cheok,
National University of Singapore (NUS)

Broadcast in CNN

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Augmented Reality Magic

Augmented Reality Magic

- Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- Along comes a pasteboard prestidigitator who folds augmented reality into the magic world

Augmented Reality Magic (card trick)

- Blend of technology and illusion creates the magic of tomorrow, today

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Marco Tempest,
Newmagic Communications Inc.,
United States

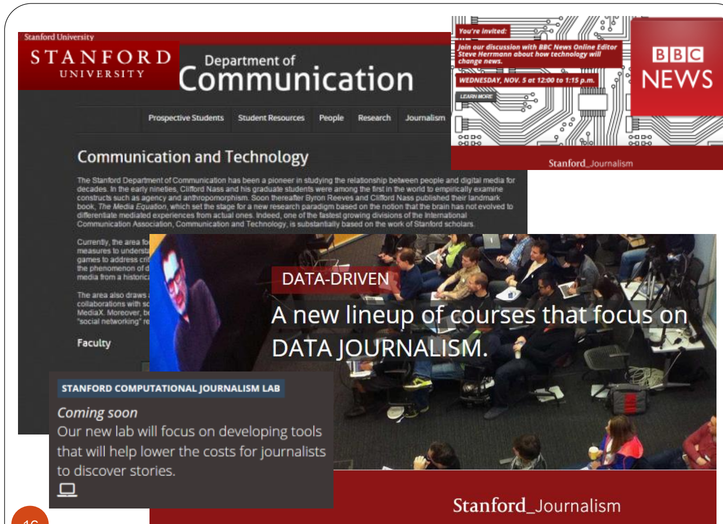
Integration leads to Innovation

Information Technology Research (Eye Tracking) + Comm. Arts Research (Advertising Strategy)

=

Research Innovation

Graduate School of Communication Arts and Management **Innovation**



http://comm.stanford.edu/technology/

Oxford Internet Institute
Understanding life online

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UNIVERSITY OF OXFORD

DPhil in Information, Communication and the Social Sciences

The DPhil in Information, Communication and the Social Sciences (doctoral) programme provides an opportunity for highly qualified students to undertake cutting edge Internet-related research. We are looking for academically excellent candidates who display the potential and enthusiasm necessary to perform research that will make a difference -- to ask important questions and to adopt innovative methodologies and approaches for exploring those questions. Many OII doctoral students are pursuing research that will shape the development of digital networked spaces and those whose lives are affected by it.

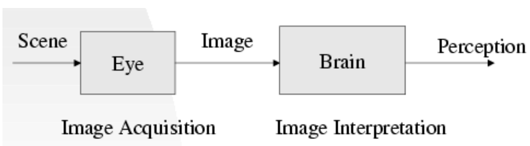

OII Research Fellow Kathryn Eccles has been appointed 'Digital Humanities Champion' for the University of Oxford. She will help develop the cross-University Digital Humanities strategy.

http://www.oii.ox.ac.uk/

Introduction to Computer Vision

Vision

- Vision is the process of discovering what is present in the world and where it is by looking.



Computer Vision

- **"Computer vision** is the science and technology of machines that see. As a scientific discipline, computer vision is concerned with the theory behind artificial systems that extract information from images. The image data can take many forms, such as video sequences, views from multiple cameras, or multi-dimensional data from a medical scanner."

By Dana H. Ballard, Computer Vision.

Computer Vision

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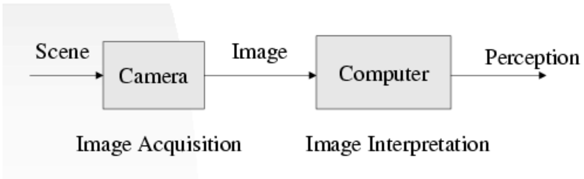

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  - Since 1960's
  - 1970's : Recognized as a technology for practical use.
  - Most Important Issues : Image Transformation
  - Noise/Blur Compensation, Distortion Correction

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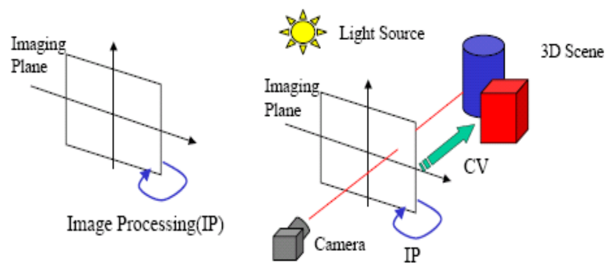
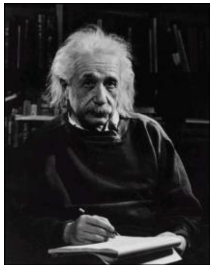
  - Computational Vision proposed by Marr
    - Shape from X(X:shading, texture, contour, motion…etc.)
    - 3D Scene Understanding from 2D images
  - Robot Vision: Vision functionality for Robots
  - Media Processing, Human Interface

Image Processing / Computer Vision



Computer Vision

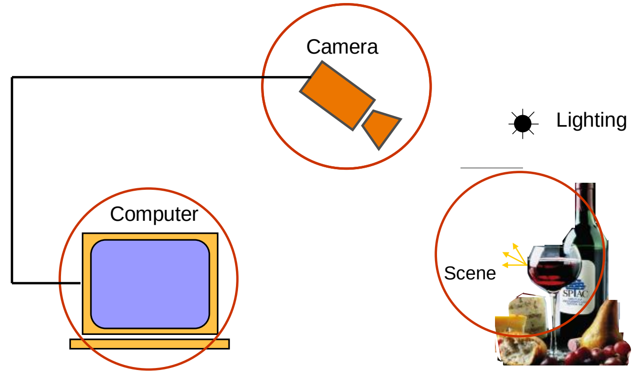


What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Components of a Computer Vision System



Scene Interpretation

Example

Finding People in images

Problem 1: Given an image I

Question: Does I contain an image of a person?

“Yes” Instances






“No” Instances

Recent Computer Vision Applications

Picture Collage

By Dr. Jian Sun
Visual Computing Group,
Microsoft Research Asia (MSRA)

What’s Picture Collage?

- Inputs:

- How to arrange a set of images on a given canvas?

Mosaic Collage

Google’s Picasa Collage

Our Picture Collage

Picture Collage

- A “visual” image summarization
  - Efficiently view a set of images
  - Maximizing visible visual information
- For image browsing, printing, and sharing
  - Causally images arrangement is easy by hand
  - But it is difficult to the user by mouse

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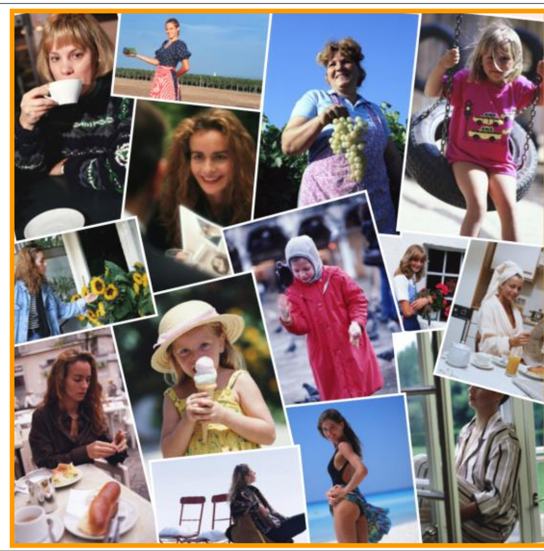

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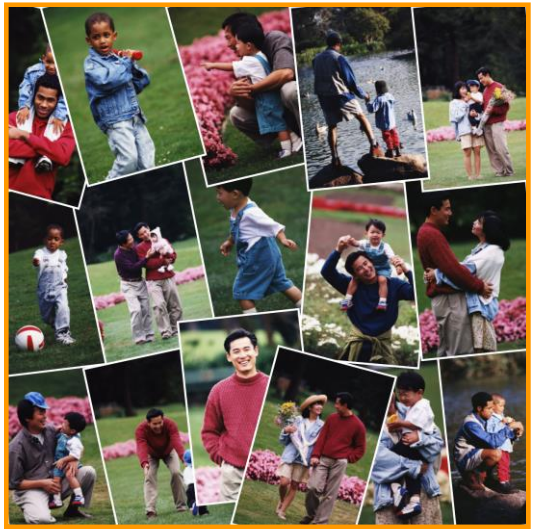

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Learning Realistic Human Actions from Movies

By Dr. Ivan Laptev
INRIA Paris, France
Institut national de recherche en informatique
et en automatique

Learning Realistic Human Actions from Movies

- The aim of this paper is to address recognition of natural human actions in diverse and realistic video settings.
- This challenging but important subject has mostly been ignored in the past due to several problems one of which is the lack of realistic and annotated video datasets.

Learning Realistic Human Actions from Movies

- Their first contribution is to address this limitation and to investigate the use of movie scripts for automatic annotation of human actions in videos.
- They evaluate alternative methods for action retrieval from scripts and show benefits of a text-based classifier.

Learning Realistic Human Actions from Movies

- Using the retrieved action samples for visual learning, we next turn to the problem of action classification in video.
- They present a new method for video classification that builds upon and extends several recent ideas including local space-time features, space-time pyramids and multi-channel non-linear SVMs.

Video result

Learning Realistic Human Actions from Movies

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By Dr. Ivan Laptev
INRIA Paris, France

Flash Matting

By Dr. Jian Sun
Visual Computing Group,
Microsoft Research Asia (MSRA)

Flash Matting

- In this paper, they propose a novel approach to extract mattes using a pair of flash/no-flash images.
- Their approach, which they call flash matting, was inspired by the simple observation that the most noticeable difference between the flash and no-flash images is the foreground object if the background scene is sufficiently distant.

- Flash matting results for flower scene with complex foreground and background color distributions.
- From left to right: flash image, no-flash image, recovered alpha matte, and flower basket with new background.

Method of Flash Matting

- They apply a new matting algorithm called joint Bayesian flash matting to robustly recover the matte from flash/no-flash images.
- Even for scenes in which the foreground and the background are similar or the background is complex

Video result

Flash Matting

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By Dr. Jian Sun from Microsoft

Drag-and-Drop Pasting

By Dr. Jian Sun
Visual Computing Group,
Microsoft Research Asia (MSRA)

(a) (b) (c) (d) (e)

- Given a source image (a), the user draws a boundary that circles the wood log and its shadow in the water, and drags and drops this region of interest onto the target image in (b).
- The result from Poisson image editing (c) is however not satisfactory. Structures in this target image (e.g., the dark beach) intersect the source region boundary, thus produce unnatural blurring after solving the Poisson equations.
- Their approach, called *drag-and-drop pasting, computes an optimized boundary shown in (d),* which is then used to generate a seamless image composite (e).

Video result

Drag-and-Drop Pasting

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By Dr. Jian Sun from Microsoft

Recent Sketching Interface Applications in CG

Dr. Takeo Igarashi,
University of Tokyo, Japan

Teddy: A Sketching Interface for 3D Freeform Design

- By Dr. Takeo Igarashi, University of Tokyo, Japan
- Teddy is a sketch-based 3D modeling software. You can make interesting 3D models just by drawing freeform strokes.

Teddy (cont.)

- Teddy requires Java installed in your machine, and mainly designed for Windows.
- Commercial Products based on Teddy: PC package , GameCube game , PlayStation2 game

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Dr. Takeo Igarashi,
University of Tokyo

Human Detection

Dr. William R. Schwartz,
University of Maryland, United States

Human Detection

- Human Detection Using Partial Least Squares Analysis
- Detect the pedestrians robustly

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Dr. William R. Schwartz,
University of Maryland,
United States

KEIO MEDIA DESIGN.
慶應義塾大学大学院メディアデザイン研究科
GRADUATE SCHOOL OF MEDIA DESIGN, KEIO UNIVERSITY
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Keio University

General Info

Digital media is changing from conventional mass media to personal media that uses media infrastructure for community building. Examples include the convergence of broadcasting and communications and the emergence of social networking services. Content adds value to media, and design enhances all aspects of life. These have become important engines in enriching the human heart and mind. For media, content, and design to take root in society, it is crucial to understand how they function as industries, to understand their business and management models, and to include policies about intellectual property a describe in "Our philosophy of education, " we train "media innovators," defined as peop

Graduate School of Media Design

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GRADUATE SCHOOL OF MEDIA DESIGN, KEIO UNIVERSITY

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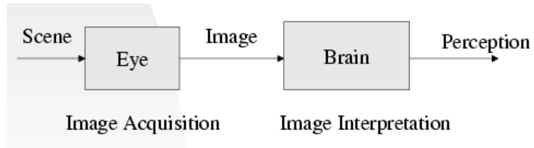

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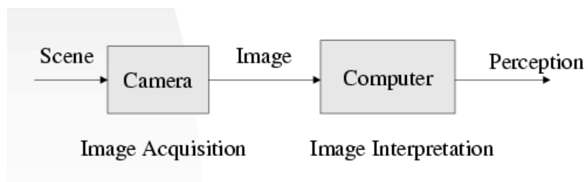

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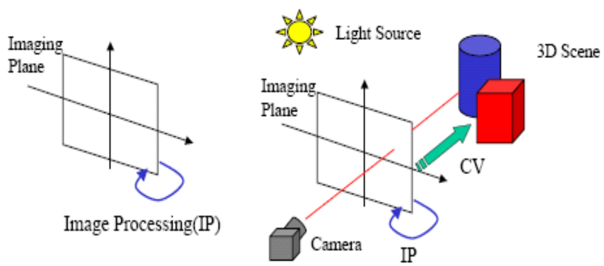

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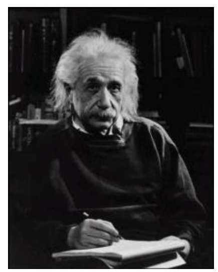

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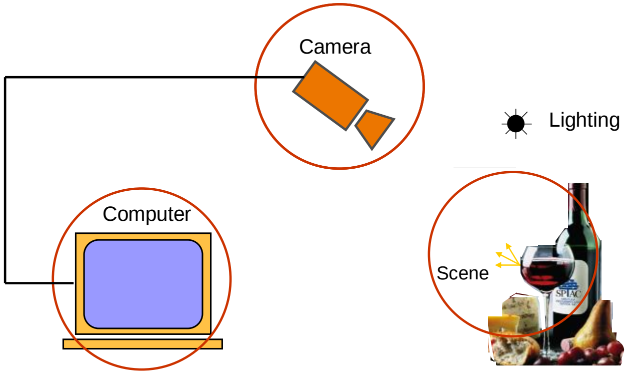

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Thank you

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